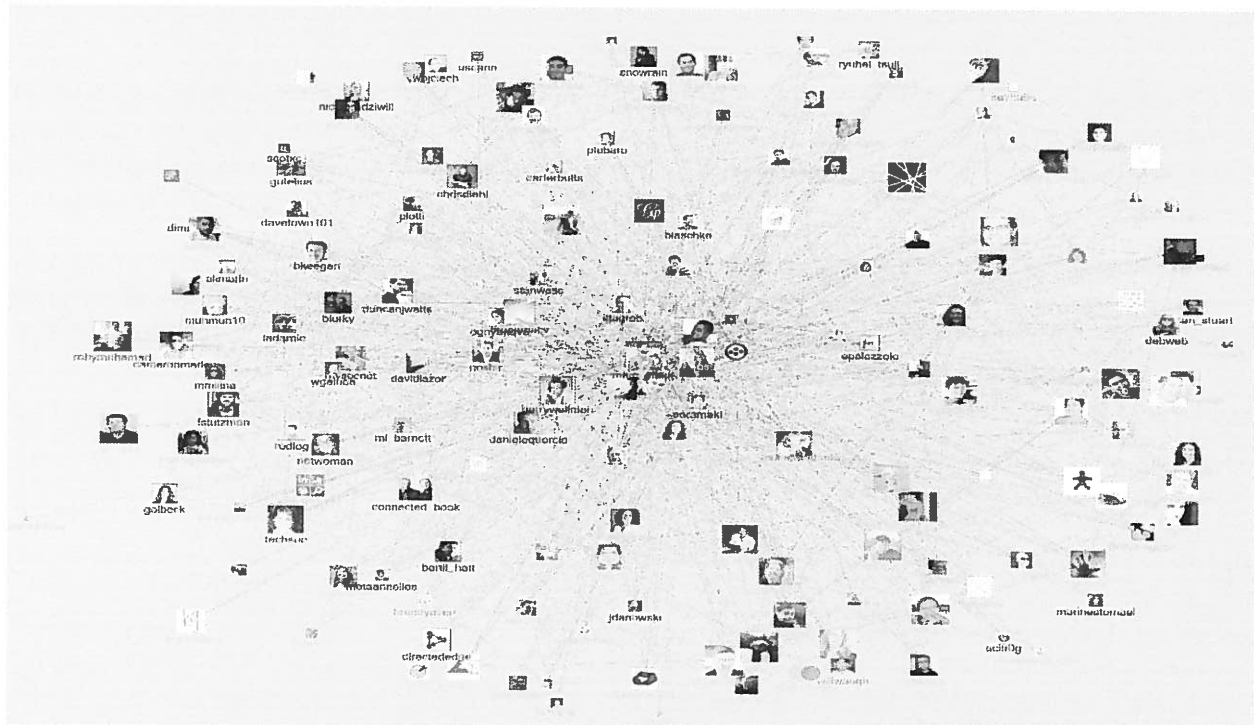


Brecksville Broadview Heights City Schools

Technology Road Map



A Technology Road Map is a living document mapping out technology needs for an organization. It is used for communication within the technology group and between the technology group and interested stake holders to illuminate priorities, inform budgets, and gauge status.

The Technology Road Map is a strategic document in that it needs to be updated as priorities change, new needs are uncovered, and budget issues are addressed. This is typically handled on a quarterly basis. All interested parties must be involved and agree on organizational priorities.

Presented by TTG, Inc.

Updated November 27, 2019

TTG has been engaged in various projects with the Brecksville Broadview Heights City School District for about 18 months. In the time frame, TTG has spent significant time in understanding the basis of the network. On behalf of the District we have bid, awarded, and implemented a new VoIP telecommunication system upgrade, reviewed and helped remediate the redesign of the network routing topology, and are currently engaged in trouble shooting and resolving a problem with the POS systems supporting food services dropping off the network.

The District has made a significant investment in identifying and resolving its network issues to date. TTG completed a multi week evaluation of the network and its readiness for VoIP (\$4,500). As a result, we provided a list of concerns for the district that needed addressed – some critical to the voice project – others not. TTG was hired to see to the remediation of the critical items for the VoIP implementation (\$76,000) but many items were identified and as needed for the network, but no work was completed towards those goals. They included wireless updates, network monitoring, alerting, as well as written policies for upgrades and services documentation. We further recommended a review of the security policies and tools.

I tell you all this because what you have requested is a discovery and remediation report. Our team has done that, been in the network, and understands how it works and its overall shortcomings. We do not recommend a formal assessment, as it's not necessary for TTG to begin to help identify and begin remediation recommendations on behalf of the District.

I have attached a copy of the report prepared last year for your review. SOW 1 and 2 have been completed and the results of a quality voice capable network are evident. The POS issues, which seemed easy to point at the network because of previous issues, are not the result of your network, but rather have been isolated to problems on your providers servers that they are working on with our guidance. SOW 3 was not done. This is the approach we recommend for what you need to ensure the technology at the new elementary school will work effectively on the District's network.

TTG would propose to put a technology professional from TTG onsite with District staff for 12 to 15 hours per week for at least the next 12 months. At that time outstanding items can be identified and hours if needed adjusted. Whenever needed we can back down the hours to a lower amount or simply end the engagement. Work and deliverables to include:

Minimum 12 hours on site/wk plus hours to support efforts Bi-Weekly update reporting	Lead technology/construction meetings to insure items not overlooked and are planned for	Coordinate efforts with the city on security design for shared space and network connectivity
Review technologies for new schools - sign off on readiness or produce remediation options	Produce WiFi testing plans and carry out for HS, MS, and District offices (elementary?) – Provide remediation recommendations	Review current infrastructure (switches, AP's, servers, and software) best practice integration

Review and provide network monitoring and alerting options	Direct and review the development of IT Policies	Review/develop Chromebook integration policies
Security Assessment review of firewalls	Develop and provide budget for Internet/Network redundancy	Finalize with 3 year go forward strategy plan to be managed by district personnel

The above items are just to provide a view into the breadth that this type of engagement can provide versus paying outside companies to go through expensive discovery to provide evaluations and bids for each portion. The following fees are significantly reduced from our original quote for this service. We have done this based on the ongoing partnership we have had with the district and understanding the district needs and operations already. Simply put, we have experience now that allows us to be more aggressive in this type of open-ended engagement.

TTG proposes to put one of our strategic team members (usually Michael Sokoloski) on site for a minimum of 12 hours per week. On top of that will be the additional hours to support this effort, follow up, preparation, reporting, etc. TTG's monthly fee for this will be \$8,250.

The District had requested a Network Assessment proposal that would outline the network infrastructure, design, software and security. We are happy to install software on the District infrastructure and run an assessment that will identify what is in place, levels of software, incompatibilities, security vulnerabilities etc. We will also include a Dark Web scan that looks for activity on the Dark Web that involved District staff's credentials. TTG would require 3 to 4 weeks to run the software and complete the write up associated. I have attached a copy of a report done for a different client for your review so you can see the way the software/reporting works. The software provides the information our team needs, and our team writes the remediation recommendations. TTG's cost for this would typically be \$22,000 and take 45 to 60 days, however we are familiar with the district infrastructure and are crediting \$7,500 against this total making our fee for the Network Assessment \$14,500. Again, we feel those monies could be invested in results given the work we have already done, but if there is a need for a document such as the one I have attached, we certainly can run our tools and write up the recommendations as a beginning phase of work.

Cost Summary:

Description	Costs
Recommended vCIO design and remediation	\$8,250/Mo estimate at 10 to 15 months
Network Assessment if desired	\$22,000
Network Assessment Credit based on work to date	\$7,500

I asked our Network Architect, Michael Sokoloski, who completed most of our work on the district, to provide a road map of how we would attack known items based on the information we had. You will find it on the following pages.

Brecksville Technology Road Map Items;

Note 1: Examples drawn from TTG engagements and the Network assessment for VoIP (2018) and POS troubleshooting projects.

Note 2: These are just the known challenges/projects – our recommendation is a more in-depth dive to uncover hidden issues so plans can be made to address them before they become emergencies.

- New School Technology Readiness
 - Network connectivity and support of business needs
 - Technology support for physical security; access control, video, multi-media printing and other business applications
 - Physical Network (fiber, copper)
 - Facility's needs; MDF/IDF space, racking, power and ups, cooling needs
 - Logical Network (switches, routers, connectivity to end devices)
 - Wireless
 - Other needs; communication systems, signage, etc
 - Integrations and challenges to all of these items from community/city shared space
- Wireless Technology
 - Aging system with significant controller problems
 - New district system needed
 - Technology selection, budget, installation and operation plans
- Internet Access
 - Existing access not redundant
 - Firewall and network security systems aging and being outgrown
- End User devices (District)
 - Upgrade and operations plan needed overall (windows 7 devices out of support, critical item)
 - Computer Lab upgrades, operations and optimization challenges
- End User devices (Student)
 - Chrome book security and district wide network security policy integrations
- Server Infrastructure and virtualization
 - Map and understand server dependencies
 - Optimize virtualization and redundancy
 - Storage
 - Cloud optimizations and usage/cost control
 - Backups
- Monitoring of Infrastructure
 - Network

- Server
- Application (POS, Printing, Voice, Video, financial, etc)
- Operationalize alerting and communications
- Change management
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- New Technologies Adoption Council
 - Evaluate new systems and applications for operation at Brecksville before purchase and then the dreaded “it works everywhere else is uttered ever again”
- Operations and Staffing Plan for Technology team
 - Map processes and work efforts for technology tasks and evaluate skill levels and staff requirements
 - Evaluate staff and 3rd party options for each work effort

Note 3; Again, these are just the examples known from past targeted evaluations and projects. There are significant gray areas currently which will add to the list.

This completes our proposal to the District. We look forward to continuing to be of service and partnering with the district on their IT Technology needs.